

TRAFFIC INTELLIGENCE SYSTEM AI-DRIVEN TRAFFIC & RISK ENGINE

*COMPARATIVE STUDY OF MACHINE LEARNING
FOR TRAFFIC VOLUME AND ACCIDENT RISK PREDICTION
PROJECT DOCUMENTATION*

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Traffic Intelligence System Documentation

PROJECT INFORMATION

Project Title	Traffic Intelligence System
Project Type	Machine Learning Project
Domain	Artificial Intelligence & Smart City Analytics
Dataset Source	Custom 50,000-Row Traffic Corpus
Algorithm Used	Linear Regression & Logistic Regression
Programming Language	Python
Frontend	HTML, CSS, JavaScript
Backend	FastAPI / Flask
Database	Serialized Model Files (.pkl)
Visualization	Matplotlib
Team Members	A. Rohit

Traffic Intelligence System is a machine learning-based engine designed to predict traffic volume and accident risk based on environmental factors like time, weather, and road quality. The system uses traditional machine learning baselines to determine the risk. In addition, the application provides a live interactive web dashboard.

ABSTRACT

The Traffic Intelligence System is an ML-based prediction engine developed using Python, Scikit-Learn, Flask, and JavaScript. The project addresses the challenge of traffic management and safety by predicting traffic volume and evaluating accident risk.

The system takes environmental conditions as input (e.g., weather, road quality, time) through a web-based dashboard, and the trained models generate the most likely traffic volume and risk level in real time.

To determine the most effective approach, two lightweight baseline models -- Linear Regression and Logistic Regression -- were trained using Scikit-Learn on a custom 50,000-row dataset. The project demonstrates how machine learning can be applied to urban planning and safety problems and supports research into predictive city systems and human-computer interaction.

RESULTS AND OUTPUTS

The live dashboard, built with HTML/JS and served through a Flask backend, allows users to select environmental conditions and view predictions from the trained models in real time.

Below is a screenshot of the Website UI successfully producing outputs based on the machine learning models:

Time of Day (Hour: 0-23)

13

Temperature (°C)

24

Weather Condition

Cloudy

Is it the Weekend?

Yes (Weekend)

Road Quality

Poor

Predict Traffic & Risk

Prediction Results:

 Expected Traffic Volume: **1,171** vehicles

 Accident Risk: **LOW RISK**